

STAT 516 - Homework 4 Answer Key

1. (9 points) A discrete random variable X has probability mass function

$$P(X = x) = \begin{cases} p, & x = 0, \\ 2p, & x = 2, \\ 3p, & x = 4, \\ 1 - 6p, & x = 6. \end{cases}$$

You are given that $E[X] = 3.6$. Determine p .

$$E[X] = 0(p) + 2(2p) + 4(3p) + 6(1 - 6p).$$

$$3.6 = 4p + 12p + 6 - 36p = 6 - 20p.$$

$$20p = 2.4$$

$$p = 0.12.$$

2. (9 points) A discrete random variable X takes values 1, 2, 4, 5 with probabilities

$$P(X = 1) = 0.10, \quad P(X = 2) = 0.25, \quad P(X = 4) = 0.40, \quad P(X = 5) = 0.25.$$

Calculate $E[X]$ and $\text{Var}(X)$.

$$E[X] = 1(0.10) + 2(0.25) + 4(0.40) + 5(0.25) = 3.45.$$

$$E[X^2] = 1^2(0.10) + 2^2(0.25) + 4^2(0.40) + 5^2(0.25) = 13.75.$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 13.75 - (3.45)^2 = 1.8475.$$

3. (10 points) Let X be a continuous random variable with density function

$$f(x) = \begin{cases} cx^2, & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Find c , and then calculate $E[X]$.

Since $f(x)$ is a density function,

$$\int_0^2 cx^2 dx = 1.$$

$$c \left[\frac{x^3}{3} \right]_0^2 = 1$$

$$c \cdot \frac{8}{3} = 1, \quad c = \frac{3}{8}.$$

Now,

$$E[X] = \int_0^2 x \cdot \frac{3}{8} x^2 dx = \frac{3}{8} \int_0^2 x^3 dx.$$

$$E[X] = \frac{3}{8} \left[\frac{x^4}{4} \right]_0^2 = \frac{3}{8} \cdot 4 = \frac{3}{2}.$$

4. **(9 points)** Let X be a continuous random variable with density function

$$f(x) = \begin{cases} \frac{2|x|}{13}, & -2 \leq x \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate $E[X]$.

$$E[X] = \int_{-2}^3 x \frac{2|x|}{13} dx.$$

Since $|x| = -x$ for $-2 \leq x < 0$ and $|x| = x$ for $0 \leq x \leq 3$,

$$E[X] = \int_{-2}^0 x \frac{2(-x)}{13} dx + \int_0^3 x \frac{2x}{13} dx.$$

$$= \frac{2}{13} \left(\int_{-2}^0 -x^2 dx + \int_0^3 x^2 dx \right).$$

$$= \frac{2}{13} \left(- \left[\frac{x^3}{3} \right]_{-2}^0 + \left[\frac{x^3}{3} \right]_0^3 \right).$$

$$= \frac{2}{13} \left(-\frac{8}{3} + 9 \right) = \frac{2}{13} \cdot \frac{19}{3} = \frac{38}{39}.$$

5. **(9 points)** Let X be a continuous random variable with density function

$$f(x) = \begin{cases} \frac{3}{26} x^2, & 1 < x < 3, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate $E \left[\frac{1}{X} \right]$.

$$E \left[\frac{1}{X} \right] = \int_1^3 \frac{1}{x} \cdot \frac{3}{26} x^2 dx.$$

$$\begin{aligned}
&= \frac{3}{26} \int_1^3 x \, dx = \frac{3}{26} \left[\frac{x^2}{2} \right]_1^3 \\
&= \frac{3}{26} \cdot \frac{9-1}{2} = \frac{3}{26} \cdot 4 = \frac{12}{26} = \frac{6}{13}.
\end{aligned}$$

6. **(9 points)** Let X be a continuous random variable with density function

$$f(x) = \begin{cases} \lambda^2 x + \frac{3}{2} \lambda^3 x^2, & 0 < x < \frac{1}{\lambda}, \\ 0, & \text{otherwise,} \end{cases}$$

where $\lambda > 0$. Verify that this is a valid density function.

$$\begin{aligned}
\int_{-\infty}^{\infty} f(x) \, dx &= \int_0^{1/\lambda} \left(\lambda^2 x + \frac{3}{2} \lambda^3 x^2 \right) \, dx \\
&= \lambda^2 \left[\frac{x^2}{2} \right]_0^{1/\lambda} + \frac{3}{2} \lambda^3 \left[\frac{x^3}{3} \right]_0^{1/\lambda} \\
&= \lambda^2 \left(\frac{1}{2\lambda^2} \right) + \frac{3}{2} \lambda^3 \left(\frac{1}{3\lambda^3} \right) \\
&= \frac{1}{2} + \frac{1}{2} = 1.
\end{aligned}$$

Also, since $\lambda > 0$ and $x > 0$, we have

$$\lambda^2 x + \frac{3}{2} \lambda^3 x^2 \geq 0.$$

Therefore, $f(x)$ is a valid density function.

7. **(10 points)** Let X be a continuous random variable with cumulative distribution function

$$F(x) = \begin{cases} 0, & x < 1, \\ \frac{(x-1)^2}{9}, & 1 \leq x < 4, \\ 1, & x \geq 4. \end{cases}$$

Calculate $P(X < 3)$ and find the density function $f(x)$.

$$P(X < 3) = F(3) = \frac{(3-1)^2}{9} = \frac{4}{9}$$

For $1 < x < 4$,

$$f(x) = F'(x) = \frac{2(x-1)}{9}.$$

Thus, the density function

$$f(x) = \begin{cases} \frac{2(x-1)}{9}, & 1 < x < 4, \\ 0, & \text{otherwise.} \end{cases}$$

8. **(10 points)** Let X be a continuous random variable with density function

$$f(x) = \begin{cases} \frac{1}{3}e^{-x/3}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate $E[X]$.

$$E[X] = \int_0^{\infty} x \cdot \frac{1}{3}e^{-x/3} dx.$$

Use integration by parts. Let

$$u = x, \quad dv = \frac{1}{3}e^{-x/3} dx.$$

Then

$$du = dx, \quad v = -e^{-x/3}.$$

Thus,

$$E[X] = [-xe^{-x/3}]_0^{\infty} + \int_0^{\infty} e^{-x/3} dx.$$

Since

$$[-xe^{-x/3}]_0^{\infty} = 0,$$

we have

$$E[X] = \int_0^{\infty} e^{-x/3} dx.$$

$$E[X] = [-3e^{-x/3}]_0^{\infty} = 3.$$

9. **(9 points)** You are given

$$E[X] = 6, \quad \text{Var}(X) = 16.$$

Calculate

$$E[(2X - 3)^2].$$

$$E[(2X - 3)^2] = 4E[X^2] - 12E[X] + 9.$$

Since

$$\text{Var}(X) = E[X^2] - (E[X])^2,$$

we have

$$16 = E[X^2] - 36,$$

so

$$E[X^2] = 52.$$

Therefore,

$$E[(2X - 3)^2] = 4(52) - 12(6) + 9 = 208 - 72 + 9 = 145.$$

10. **(9 points)** Let X be a random variable such that

$$E[X] = 3, \quad E[X^3] = 36, \quad E[(X - 3)^3] = 0.$$

Calculate $\text{Var}(X)$.

$$(X - 3)^3 = X^3 - 9X^2 + 27X - 27.$$

Taking expectations,

$$E[(X - 3)^3] = E[X^3] - 9E[X^2] + 27E[X] - 27 = 0.$$

$$0 = 36 - 9E[X^2] + 81 - 27.$$

$$0 = 90 - 9E[X^2].$$

$$E[X^2] = 10.$$

Therefore,

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 10 - 3^2 = 1.$$

11. **(7 points)** The annual repair cost X for a machine has mean 500 dollars and variance 900. After a price increase, the new repair cost is

$$Y = 1.15X + 20.$$

Calculate $\text{Var}(Y)$.

$$\text{Var}(aX + b) = a^2 \text{Var}(X).$$

Thus,

$$\begin{aligned} \text{Var}(Y) &= (1.15)^2(900). \\ &= 1.3225(900) = 1190.25. \end{aligned}$$